Worksheet 4-6 A Local Linearity and Linearization #1

AP Calculus Name_____

Do not use a calculator unless you are told the question is calculator active.

1. Let $f(x) = 2\cos(3x) + 5$. Write an equation of the line tangent to $f(x)$ at $= \frac{\pi}{6}$. Use the tangent line to approximate $f(1)$.

2. The function g is differentiable with $g(3) = -4$ and $g'(3) = 2$. Use the tangent line to g at $x = 3$ to approximate $g(3.1)$.

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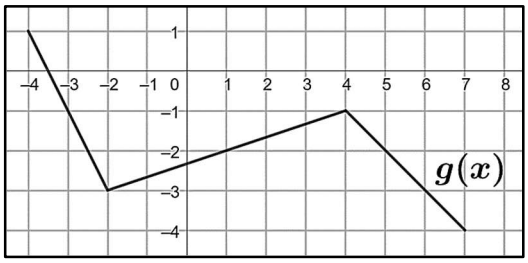
x	-3	0	1	5
$f(x)$	2	-3	6	9
$f'(x)$	4	5	3	-24



For 3 – 6, the function $f(x)$ is differentiable with selected values of f and f' given in the table above. The graph of g consists of three line segments as shown in the figure above.

3. Let $h(x) = f(x)g(x)$. Use the tangent line to h at $x = 1$ to approximate $h(2)$.

4. Let $k(x) = g\big(f(x)\big)$. Write an equation of the line tangent to $k(x)$ at $x = 0$. Use the tangent line to approximate $k(-0.3)$.

5. Let $m(x) = \sqrt{f(x)}$. Use the tangent line approximation to $m(x)$ at $x = 5$ to approximate $m(5.1)$.

6. Let $p(x) = f\big(g(x)\big)$. Approximate $p(6.2)$ by using a linear approximation for $p(x)$ at $x = 6$.

7. The function L is differentiable with a derivative defined as $\frac{dL}{dt} = \frac{1}{3}(12 - L)$. It is known that $L(9) = 6$. Write an equation for the line tangent to the graph of L at $t = 9$. Use this equation to approximate L at $t = 7$.

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8. The bottom of a cylindrical tank filled with water is leaking. The rate of change of volume (V) of water in the tank at time t is given by $\frac{dV}{dt} = -2\sqrt{V}$. At time $t = 0$, $V = 81$. Use a linear approximation for V at $t = 0$ to approximate the volume of water in the tank when $t = 4$.

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t (minutes)	0	1	3	7
$E'(t)$ (people/minute)	8	6	11	9

9. For $t \geq 0$ minutes, the number of people that have entered a department store is modeled by the function E. At $t = 1$, a total of 26 people have entered the department store. Write an equation of the line tangent to E at $t = 1$. Use this equation to approximate the total number of people that have entered the department store at time $t = 2$.

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10. **(Calculator Active)** After a large snowstorm, snow is plowed into a large mound to the side of a parking lot. As the snow melts, the amount of snow remaining can be modeled by $S(t) = 562e^{-0.5t}$ where t is measured in days. After 7 days, the amount of snow remaining is better estimated by the linear approximation to S at $t = 7$ days. Use this linear approximation to S to estimate the amount of snow remaining at $t = 8$ days.